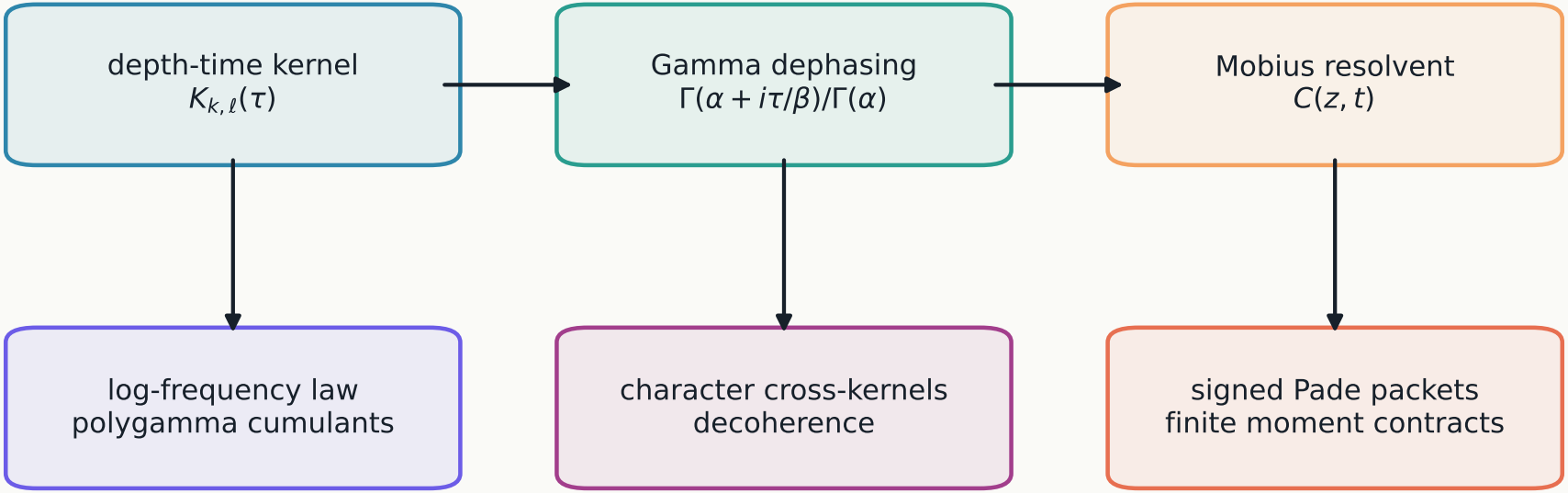Function family generated in Paper IV